Week 5 Day 5| Apache Spark

Scala | map() method

A **collection** in Scala is a data structure which holds a group of objects. Examples include Arrays, Lists, etc.

map() widely used method for **transformation** of these collections.

map() method:

- is a higher-order function
- takes another function as a parameter
- applies the function to every element in the source collection
- returns a new collection of the same type as the source collection

difference between map() and flatmap()

map() transforms a collection by applying function to each element of that function.

flatmap() combines mapping and flattening. The flatten method will collapse the elements of a collection to create a single collection with elements of the same type.

Below is an example of using map() method:

```
scala> val rdd= sc.parallelize(Seq("Today it is a gloomy da
y!", "Studying Apache Spark."))
val rdd: org.apache.spark.rdd.RDD[String] = ParallelCollectio
nRDD[6] at parallelize at <console>:1

scala> val rdd_split= rdd.map(x=>x.split(" "))
val rdd_split: org.apache.spark.rdd.RDD[Array[String]] = MapP
artitionsRDD[7] at map at <console>:1
```

```
scala> rdd_split.collect()
val res7: Array[Array[String]] = Array(Array(Today, it, is,
a, gloomy, day!), Array(Studying, Apache, Spark.))
```

Example of using flatmap():

```
scala> val rdd_split_flatmap= rdd.flatMap(x=>x.split(" "))
val rdd_split_flatmap: org.apache.spark.rdd.RDD[String] = Map
PartitionsRDD[8] at flatMap at <console>:1


scala> rdd_split_flatmap.collect()
val res9: Array[String] = Array(Today, it, is, a, gloomy, da
y!, Studying, Apache, Spark.)
```